Modeling Student Interactions in Online Statistics Courses

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ABSTRACT: This study examined collaborative assignments in an online asynchronous introductory statistics course. Written interactions from 33 undergraduate students were analyzed over a 10-week period. Social presence over time showed more variation in interactions across groups than within groups. Agreement, self-disclosure, and asking questions are the most used indicators across all groups and CKs. The correlation coefficient between total social presence and final course scores indicate a weak positive relationship.

COLLABORATIVE KEYS (CKs)

CKs are collaborative assignments given to groups of 3-4 students. Each group has a shared document that contains statistical questions. The document is organized into three sections: initial answers, discussion, and final group answer (Sabbag et al., 2025).

COMMUNITY OF INQUERY FRAMEWORK (CoI)

To explore students' interactions in the CKs, we adopted the CoI (Garrison et al., 1999). Using the CoI, we measure *Teacher Presence*, *Social Presence* and *Cognitive Presence* in students' answers.

- **Teacher Presence (TP):** reference to the instructor, videos, and materials.
- **Social Presence (SP):** Affective (expressing emotion, using humor, and self-disclosure); Interactive (quoting others, asking questions, answering questions, expressing agreement, and expressing appreciation); and Cohesive (using vocatives, using group pronouns, and phatics).
- **Cognitive Presence (CP)**: Triggering Events (recognizing the problem, sense of puzzlement; Exploration (self-correction/evaluation, divergence within the online community, divergence within a single message, information exchange, suggestions for consideration, brainstorming, leaps to conclusions; Integration (convergence among group members, convergence within a single message, connecting ideas, peer-evaluation, creating solutions; Resolution (testing solutions, defending solutions).

EXAMPLE OF COLLABORATIVE KEY

Question: Compute the value of the t-statistic for your class data. Show your work.

INITIAL ANSWERS

Student W:
t = (class average - average)/SD
= (6.87-8)/0.186 = -6.07

Student B:
t = 8.005-6.87/0.182/√70
= 41.80

Student G
$\bar{x}$ = 6.87, μ0 = 8, s = 1.03, n = 70
t = (6.87-8)/(1.03 /root70) = -9.187

Student P
t = (6.87-8)/0.186
= -6.07

DISCUSSION

Student G: Wait is the standardized statistic the same as the t-statistic? I used the formula right above question 30 but I'm not sure if that's right.

➡ **Teacher**: NA
Social: self-disclosure; ask question
Cognitive: sense of puzzlement; self-evaluation and statistics ideas

Student B: I am also confused Student G. Comparing my answer to the rest, I got a significantly larger value. I did notice I made a mathematical error switching the means.

➡ **Teacher**: NA
Social: express agreement; self-disclosure; use vocatives; answer question
Cognitive: sense of puzzlement; self-evaluation and statistics ideas

Student P: I'm definitely a bit confused by the t-test, because it seemed to contradict the p-value test, so I'm not sure if I did it correctly or not.

➡ **Teacher**: NA
Social: self-disclosure
Cognitive: sense of puzzlement; self-evaluation and statistics ideas

Student G: Guys based on the wrap up video I think the formula I used was right, Professor Sabbag also got -9.18. Is it ok if we put that down? I think you guys used the z-score formula.

➡ **Teacher**: reference to videos and instructor
Social: ask question
Cognitive: suggestion for consideration, connect ideas, peer-evaluation, defend solutions video

Student P: Oh, I see what I did wrong now, thanks for explaining, Student G!

➡ **Teacher**: NA
Social: self-disclosure; express appreciation; use vocatives; answer question; using phatics
Cognitive: unsubstantiated agreement

FINAL GROUP ANSWER

t = (6.87-8)/(1.03 /root70) = -9.187

REFERENCES

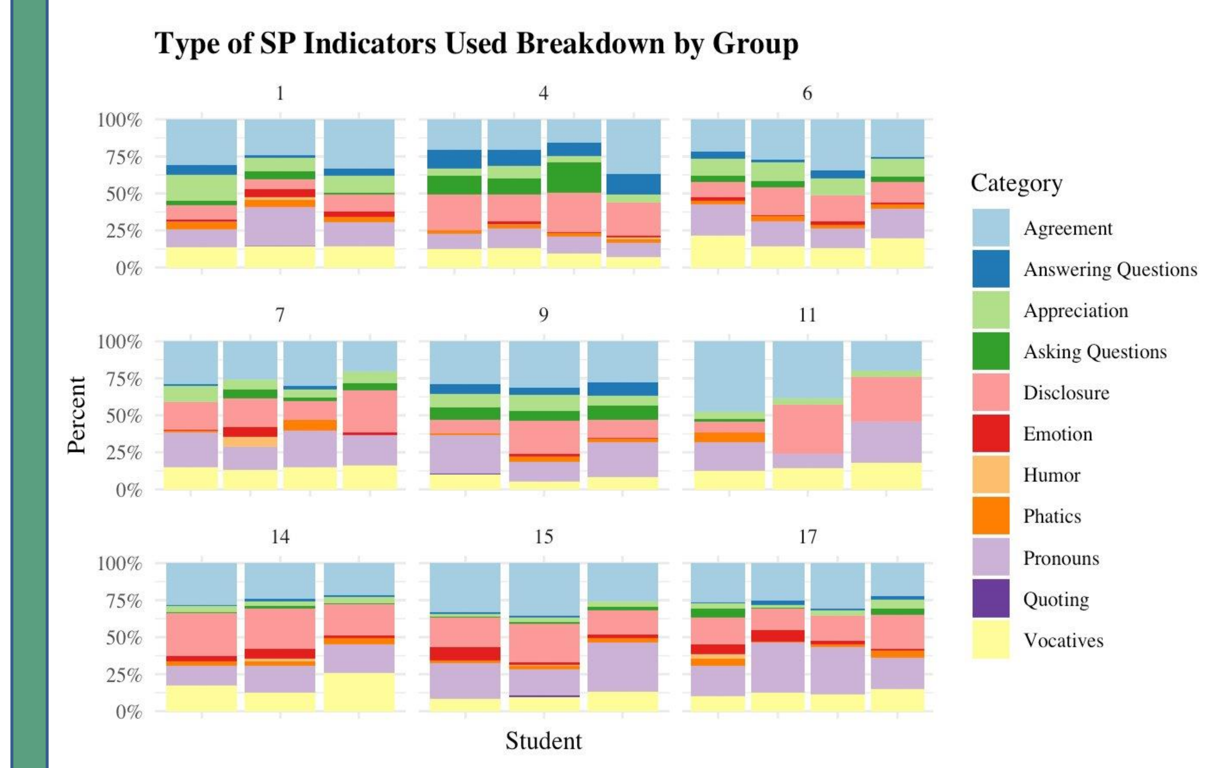
Garrison, D. R., Anderson, T., and Archer, W. (1999), "Critical Inquiry in a Text-Based Environment: Computer Conferencing in Higher Education," *The Internet and Higher Education*, *2*, 87–105.

Rourke, L., Anderson, T., Garrison, D. R., & Archer, W. (1999). Assessing social presence in asynchronous text-based computer conferencing. *The Journal of Distance Education, 14*(2), 50–71.

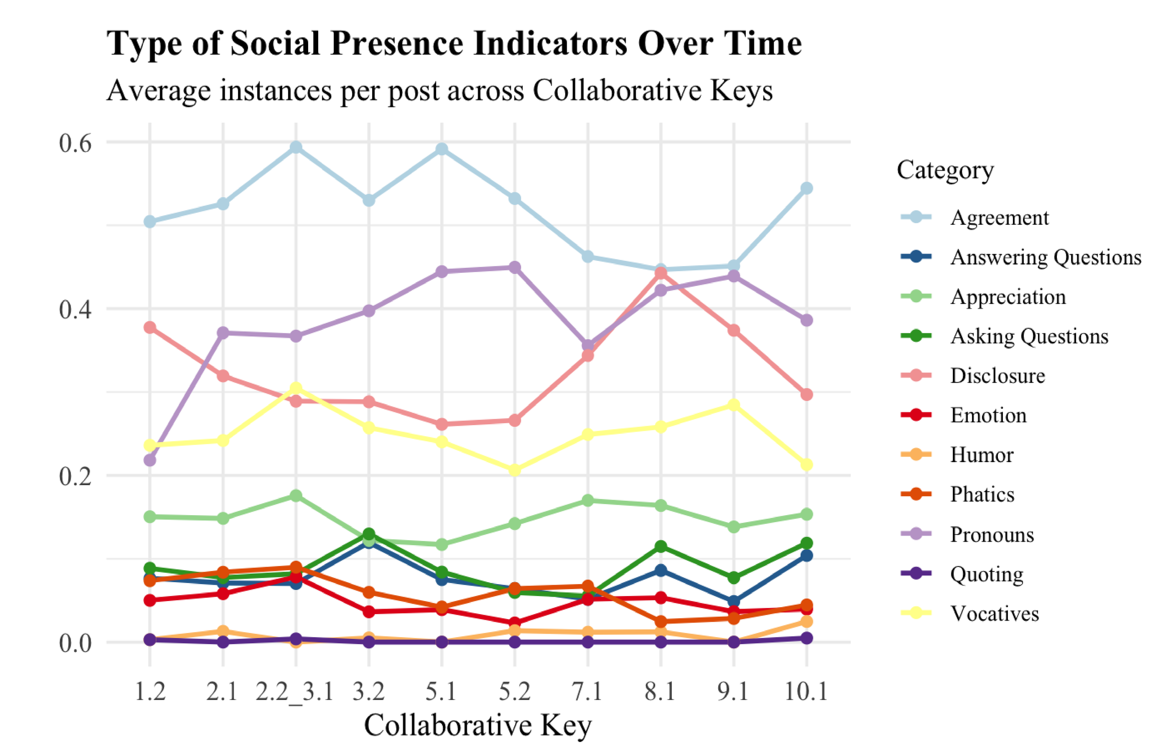
Sabbag, A., Frame, S., Justice, N., Laundroche, L., & Roggenkemper, R. (2025). The Development of Collaborative Keys to Promote Engagement in Undergraduate Online Asynchronous Statistics Courses. *Journal of Statistics and Data Science Education*, *33*(2), 152–165.

SOCIAL PRESENCE INDICATORS

The following graph reveals how students use different social presence indicators within and between groups. Agreement between group members appears to be the highest category of social presence for most students and groups. In groups 4 and 9, there is a higher proportion of students both asking and answering questions. Additionally, self-disclosure (vulnerability) was a commonly used form of social presence across the groups.

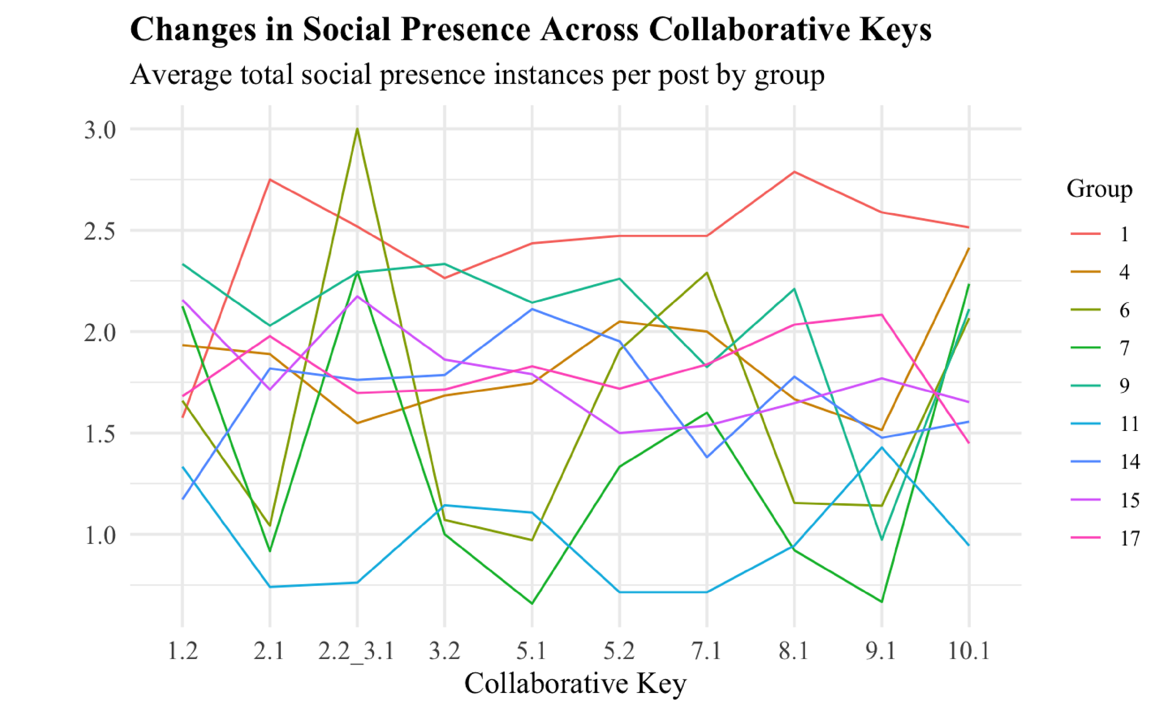


The following graph shows the average number of instances each indicator appears per post for each CK. Agreement is the category with the highest average across all CKs, but it tends to decrease after the first midterm (CK 5.1). Pronouns, self-disclosure, and use of inclusive pronouns also show relatively higher averages, however average instances of vulnerability displayed in CKs tends to increase after the first midterm.

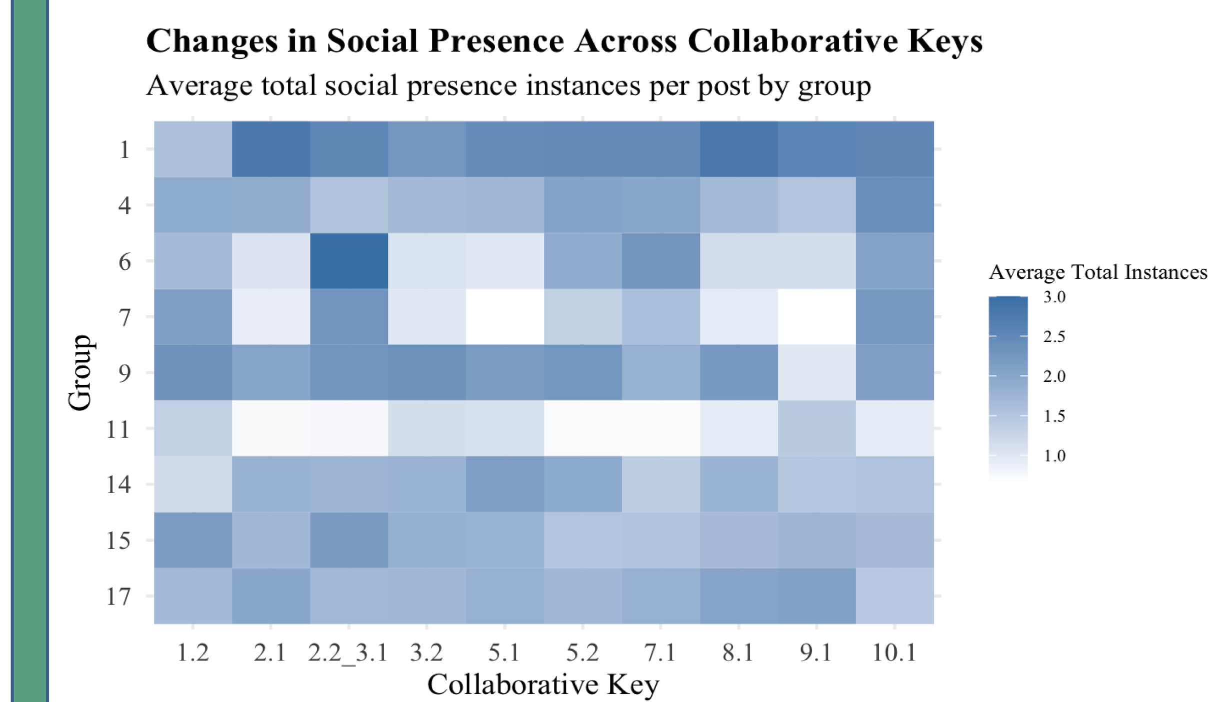


SOCIAL PRESENCE OVER TIME

The line graph and heatmap in the column to the right illustrate how each group's average total social presence (across all categories) changes over time. Each plot shows the average total instances of social presence per post for each group at each CK. Most groups show minor fluctuations with averages ranging between about 1.5 and 2.5 total instances per post. However, some groups, particularly Groups 6 and 7, display more variability, with their average total social presence ranging from below 1 to around 2.5 instances per post across different CKs. Group 1 shows consistently higher levels of total social presence throughout the entire quarter, with an average of 2.4 total social presence instances per post across all the CKs. In contrast, Group 11 shows the lowest levels, with average of 0.99 total social presence instance per post across the CKs.

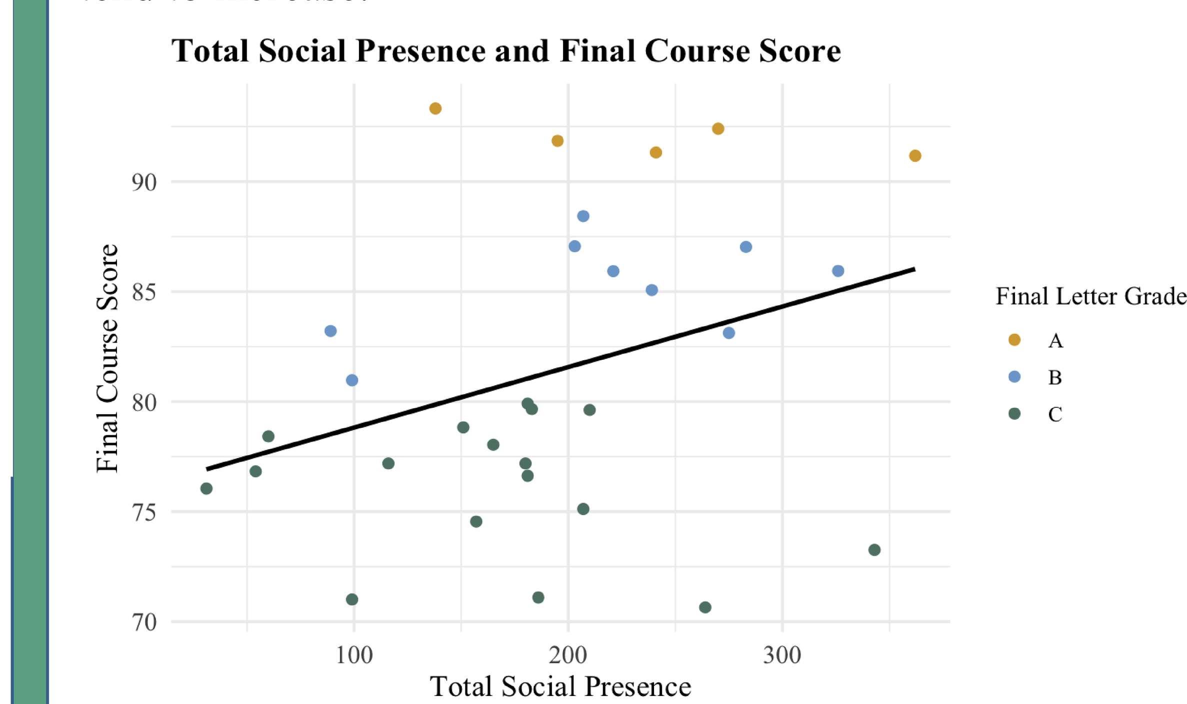


The heatmap below reflects the same patterns as the graph above through color intensity, highlighting relatively consistent average total social presence within groups and more noticeable differences in these averages across groups.



PERFORMANCE

The scatterplot shows a weak positive relationship between total social presence and final course score (r = 0.33). As total instances of social presence increase, final course grade also tend to increase.



FURTHER RESEARCH

- Investigate the use of Artificial Intelligence to help categorize students' answers.
- Continue coding and investigating Cognitive Presence and its relationship with course performance.

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